

BAYESIAN CLASSIFICATION, ESTIMATION AND PREDICTION OF GROWTH CURVES

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SUMMARY

In this paper a generalization of the work of Lee (1977) on the growth curve model with respect to the classification of an observed matrix V is attempted. Various assumptions on the parameters are considered as well as two covariance structures. Partial and conditional prediction are also considered. The approach is Bayesian.

1. Introduction

The growth curve model was introduced by Potthoff and Roy (1964) as:

$$(1.1) \quad E(Y_{p \times N}) = X_{p \times m} \tau_{m \times r} A_{r \times N}$$

where τ is unknown and X and A are known matrices of ranks $m < p$ and $r < N$, respectively.

The columns of Y are independently normally distributed with common covariance matrix Σ , i.e.

$$(1.2) \quad G(Y|\tau; \Sigma) = N[\cdot; X\tau A; \Sigma \otimes I_N]$$

The situation where c growth curves

$$(1.3) \quad G(Y_i|\tau_i; \Sigma_i; \pi_i) = N[\cdot; X\tau_i A_i; \Sigma_i \otimes I_{N_i}] \quad i = 1, 2, \dots, c$$

were considered was studied by Lee (1977). Lee studied the problem of the classification of a future set of observations V , of dimension $p \times K$, which is known to be drawn from

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one of the c normal populations, $\pi_1; \pi_2; \dots; \pi_c$ with prior probabilities $q_1; q_2; \dots; q_c$, i.e.

$$(1.4) \quad G(V|\tau_i; \Sigma_i; \pi_i) = N[\cdot; X\tau_i F_i; \Sigma_i \otimes I_K].$$

F_i is a known matrix, usually formed by some columns of A_i for $i = 1, 2, \dots, c$. For the case where q_i is assumed unknown, Lee (1977) referred to Geisser (1964).

For the purpose of classification the predictive density of V , $g(V|Y; \pi_i)$ must be derived, where Y represents all observed data $Y_1; Y_2; \dots; Y_c$. The predictive probability that V belongs to π_i is;

$$(1.5) \quad \Pr(V \in \pi_i | Y; q) \propto q_i g(V|Y; \pi_i)$$

for $i = 1, 2, \dots, c$ and $q = (q_1; \dots; q_c)$.

Lee (1977) considered the derivation of the predictive density of V , under two different covariance structures:

(i) Σ arbitrary positive definite and

(ii) Rao's simple structure: $\Sigma = X\Gamma X' + Z\Theta Z'$ where Z is $p \times (p-m)$ such that $X'Z = 0$.

In considering this predictive densities, Lee derived some of the densities under special cases ($c=2$; $r=1$). Lee also considered the matrix τ of dimension $m \times r$ for all $i = 1, \dots, c$. This means that all the populations have the same number of subdivisions. The purpose of this paper is to generalize the special cases in the paper of Lee as well as to reconsider the theory w.r.t. general r_i ($i=1, \dots, c$). Where the exact distribution could not be found, the limiting distribution was found.

A Bayesian approach to the derivation of the predictive density of V and the estimation of the parameters were used. The estimation of the parameters w.r.t. the special cases of Lee (1977) are generalized.

Another problem of great practical importance is that it must be possible to classify any submatrix of V , say $V^{(1)}$, instead of the whole of V . This can be achieved by obtaining the marginal predictive density of a submatrix of V . (Lee and Geisser (1972; 1975)). In this paper the marginal predictive density of $V^{(1)}$ as well as the conditional predictive density of any other submatrix $V^{(2)}$ given $V^{(1)}$, can be obtained directly from the generalized results.

2. Arbitrary positive definite Σ .

2.1 $\Sigma_i = \Sigma$ but unknown; τ_i unknown.

Lee (1977) discussed this case in detail but it is reconsidered in this section with respect to general r_i , i.e. τ_i is of dimension $m \times r_i$.

By using the prior density:

$$g(\tau_1; \dots; \tau_c; \Sigma^{-1}) \propto |\Sigma|^{(p+1)/2}$$

it is easy to see that, a posteriori:

$$(2.1) \quad G(\Sigma^{-1} | \tau, Y) = W \{ \cdot; [(Y - X\tau A)(Y - X\tau A)']^{-1}; N \} \text{ and}$$

$$(2.2) \quad G(\tau | Y) = D_{m,r}(\cdot; \hat{\tau}; G; (X'S^{-1}X)^{-1}; N)$$

$$\text{where} \quad r = \sum_{i=1}^c r_i; N = \sum_{i=1}^c N_i; \quad Y = (Y_1; \dots; Y_c);$$

$$\tau = (\tau_1; \dots; \tau_c) \text{ and } A = \text{diag}(A_1; \dots; A_c).$$

$\hat{\tau}$, S and G are defined in Lemma 1 of Lee (1977).

From (2.1) and (2.2) it follows that:

$$(2.3) \quad E(\tau | Y) = \hat{\tau}$$

$$(2.4) \quad E(\Sigma | Y) = (N-p-1)^{-1} \{ (Y - X\hat{\tau}A)(Y - X\hat{\tau}A)' + (N-m-r-1)^{-1} X(X'S^{-1}X)^{-1}X' (\text{tr}G^{-1}AA') \}$$

The predictive density of V can easily be obtained as:

$$(2.5) \quad g(V | Y; \tau_i) \propto |G_i|^{m/2} \cdot |Z'(YY' + VV')Z|^{-\frac{1}{2}(N+K-m)} \cdot |(X'S^{-1}X)^{-1} + B(V - XQ_i)G_i(V - XQ_i)'B'|^{-\frac{1}{2}(N+K-r)}$$

$$\text{where} \quad G_i^{-1} = M_i^{-1} + (V - \hat{V}_i)'Z(Z'S_iZ)^{-1}Z'(V - \hat{V}_i)$$

$$M_i = I - F_i^{*'}(H_iH_i')^{-1}F_i^*; H_i = (A; F_i^*)$$

$$\hat{V}_i = YA'(AA')^{-1}F_i^*; Q_i = B\hat{V}_i + BSZ(Z'SZ)^{-1}Z'(V - \hat{V}_i)$$

$$S = Y(I - A'(AA')^{-1}A)Y'; B = (X'X)^{-1}X' \quad \text{and}$$

$$F_i^* = \begin{bmatrix} 0 & \dots & 0 \\ \cdot & \dots & \cdot \\ 0 & \dots & 0 \\ & F_i & \\ 0 & \dots & 0 \\ \cdot & \dots & \cdot \\ 0 & \dots & 0 \end{bmatrix} \left\{ \begin{array}{l} \sum_{j=1}^{i-1} r_j \text{ rows} \\ \\ \sum_{j=i+1}^c r_j \text{ rows.} \end{array} \right.$$

Since the marginal and conditional predictive densities of any submatrix of V cannot be obtained from (2.5), the following limiting density was obtained: If $N \rightarrow \infty$ the predictive density of V given in (2.5) can be approximated by:

$$(2.6) \quad g(V|Y; \tau_i) \propto \text{etr} \{ -\frac{1}{2} [(\nu - \hat{w})' \{ (N+K-m)I_K \otimes Z(Z'YY'Z)^{-1}Z' \} (\nu - \hat{w})] \}$$

where ν is defined from V similarly to β in Eq. 8.6.13 of Press (1972) and $\hat{w}$ from $\hat{W}$

$$\text{where} \quad \hat{W} = (I - XBT)^{-1}XB(I - T)\hat{V}_i$$

$$\text{and} \quad T = SZ(Z'SZ)^{-1}Z'$$

Since XBT contains the stochastic matrix Y_i the probability is zero that the matrix XBT has an eigenvalue equal to one and therefore the probability is one that $(I - XBT)^{-1}$ exists.

From (2.6) it can be that, as $N \rightarrow \infty$, V has a normal distribution with mean $\hat{W}$ and covariance matrix Σ^* where:

$$\Sigma^{*-1} = (N+K-m)I_K \otimes Z(Z'YY'Z)^{-1}Z'$$

It is therefore possible to obtain the marginal as well as the conditional predictive density of any submatrix of V given some observed part of V . The marginal predictive density can be used to classify a submatrix of V . For the purpose of classification with respect to the whole of V , it would in any case be much more convenient to use (2.6) instead of (2.5).

2.2 Σ_i known; $\tau_i = \hat{\tau}$ unknown

By considering the improper prior density of the unknown parameter $\hat{\tau}$ as to be

proportional to a constant, it can be shown that the posterior distribution of $\hat{\tau}$ is given by:

$$(2.7) \quad G(\hat{\tau} | \Sigma_i (i=1, \dots, c); Y) = N \left\{ \cdot; \hat{\tau}_0^*; \left(\sum_{j=1}^c \Omega_j \right)^{-1} \right\}$$

where $\hat{\tau}_0^*$ is a matrix of dimension $(m \times r)$ formed from

$$\hat{t}_0 = \left\{ \sum_{j=1}^c \Omega_j \right\}^{-1} \left\{ \sum_{j=1}^c \Omega_j \hat{t}_{0j} \right\}$$

similarly to ν in (2.6),

$$\Omega_j = (X' \Sigma_j^{-1} X) \otimes (A_j A_j'),$$

$\hat{t}_{0j}$ is defined from $\hat{\tau}_{0j}$ in the same way as $\hat{t}_0$ and $\hat{\tau}_{0j}$ is defined similarly to $\hat{\tau}$ given in lemma 1 of Lee (1977).

From (2.7) it follows that: $E(\hat{\tau} | \Sigma_i (i=1, \dots, c); Y) = \hat{\tau}_0^*$.

By considering the special case where $c=2$ and $r=1$ these results reduce to the work of Lee (1977).

Consider the pseudo-augmented model:

$$(2.8) \quad E(Y_i) = (X_i Z)' \begin{bmatrix} A_i \\ Z_i^* \end{bmatrix}$$

where Y_i is a $p \times N_i$ matrix,

$$T(p \times N_i) = \begin{bmatrix} \tau & O \\ O & O \end{bmatrix} \begin{matrix} m \\ (p-m) \end{matrix}$$

$$r \quad (N_i - r)$$

$(X_i Z)$ and $\begin{bmatrix} A_i \\ Z_i^* \end{bmatrix}$ are square matrices partitioned such that the

multiplication $X_i \tau A_i$ is admissible,

$$X'Z = O \text{ and } A_i Z_i^{*'} = O.$$

It can then be shown that the predictive density of V is given by:

$$(2.9) \quad g(V|\Sigma_i(i=1, \dots, c); Y; \pi_i)$$

$$\alpha \text{ etr } \{-\frac{1}{2} [v - A_{22}^{-1} \{A_{22}m_{i2} - A_{21}(y_i - m_{i1})\}]' A_{22} \\ [v - A_{22}^{-1} \{A_{22}m_{i2} - A_{21}(y_i - m_{i1})\}]\}$$

where $A_{21}(pK \times pN_i)$ and $A_{22}(pK \times pK)$ are submatrices of the matrix

$$A = (DD') \otimes \Phi' A^* \Phi \text{ and } \Phi \text{ is the left g-inverse of } Z.$$

$$m_i = \left[\begin{array}{c} m_{i1} \\ m_{i2} \end{array} \right] \left\{ \begin{array}{l} pN_i \text{ elements} \\ pK \text{ elements} \end{array} \right. \quad \text{is found}$$

from the columns of $M_i = M_i^* D^-$.

$$D^- \text{ is left g-inverse of } D = \sum_{j=1}^{N_i+K} D_j \text{ and } D_j \text{ is the } j\text{-th column of } \left[\begin{array}{c} A_i \\ Z_i^* \end{array} \right]^{-1}$$

$$M_i^* = ZA^{*-1}$$

$$A^* = S_i^* + B\ell$$

$$B = S_i^* P_i [P_i' S_i^* P_i + \sum_{\substack{j=1 \\ j \neq i}}^c P_j' S_j P_j]^{-1} P_i S_i^*$$

and

$$\ell = -S_i^{*-1} P_i (P_i' P_i)^{-1} \left(\sum_{\substack{j=1 \\ j \neq i}}^c P_j' S_j \lambda_j \right)$$

$$\lambda_j = \left[\begin{array}{c} (X;Z)^{-1} (Y_i;V) D_1 \\ \dots\dots\dots \\ (X;Z)^{-1} (Y_i;V) D_{N_j} \end{array} \right]$$

$$S_i^* = (X;Z)' \Sigma_i^{-1} (X;Z) \otimes H_i \text{ where } H_i \text{ is defined similarly to } H_i \text{ in Sec}$$

2.1. S_j is defined similarly to S_i^* but without the F_i

$$\text{and } P_j = \left[\begin{array}{l} I_m \\ \{(p-m) \text{ rows containing zeros}\} \\ I_m \\ \{(p-m) \text{ rows containing zeros}\} \\ \vdots \\ p \times (N_j - r) \quad j=1, \dots, c; \quad j \neq i \\ \vdots \\ p \times (N_i + K - r) \quad j=i \end{array} \right] \quad \begin{array}{l} \text{This is repeated} \\ r \text{ times with } I_m \\ \text{moving to the right.} \\ \\ \text{rows containing} \\ \text{zero elements} \end{array}$$

This is a matrix of order $(pN_j \times mr) \quad j \neq i; \quad j=1, \dots, c; \quad (p(N_i + K) \times mr) \quad j=i, \quad i=1, 2, \dots, c.$

It is clear that (2.9) is the density of a normal distribution. Note however that in this case we have the exact density.

From (2.9) the marginal and conditional predictive densities of submatrices of V can be found from standard normal theory.

3. Rao's simple structure: $\Sigma = X\Gamma X' + Z\Theta Z'$

3.1 $\Sigma_i = X\Gamma X' + Z\Theta Z'; \Gamma, \Theta$ and τ_i unknown.

By taking the joint prior density of the unknown parameters as:

$$g(\tau_i (i=1, \dots, c); \Gamma^{-1}; \theta^{-1}) \propto |\Gamma|^{(m+1)/2} |\theta|^{(p-m+1)/2}$$

the Bayes estimations are given by:

$$\begin{aligned} E(\tau_i | Y) &= T_{1i} = Y_i A_i' (A_i A_i')^{-1} \\ (3.1) \quad E(\tau | Y) &= (N - r - m - 1)^{-1} B S B' \\ E(\Theta | Y) &= (N - p + m - 1)^{-1} D Y Y' D' \end{aligned}$$

where N, r, B, A and S have the same definitions as in Sec 3.1 and $D = (Z'Z)^{-1}Z'$.

The predictive density of V can be derived from the joint distribution of V and $\tau_i (i=1, \dots, c)$ as:

$$(3.2) \quad g(V|Y; \pi_i) \propto |D(Y Y' + V V') D'|^{-(N+K)/2} \\ \cdot |BSB' + (BV - T_{1i} F_i) M_i (BV - T_{1i} F_i)'|^{-\frac{1}{2}(N+K-r)}$$

where $M_i = I - F_i'(H_i H_i')^{-1} F_i$ and $H_i = (A_i; F_i)$.

It can be shown that if $N \rightarrow \infty$, (3.2) can be approximated by:

$$(3.3) \quad g(V|Y; \pi_i) \propto \exp \left\{ -\frac{1}{2} [(v - \hat{v})' \{ (N+K) I_K \otimes D' (D Y Y' D')^{-1} D \} (v - \hat{v})] \right\}$$

where v is formed from the columns of V and
 $\hat{v}$ is formed from the columns of $XT_{1i} F_i$

From (3.3) it can be seen that, as $N \rightarrow \infty$, V has a normal distribution with mean $XT_{1i} F_i$ and covariance matrix Σ^* where:

$$\Sigma^{*-1} = (N+K) I_K \otimes D' (D Y Y' D')^{-1} D$$

The marginal and conditional predictive densities of any submatrix of V can now be obtained from (3.3) by standard normal theory. For classification purposes with respect to the whole of V , it would, as in Sec. 2.1, be much more convenient to use (3.3) in stead of (3.2).

4. An example

The theory in Sec. 2.1 is illustrated by using the rainfall data given in Gabriel (1972). These rainfall data were measured in Israel during the rainy season i.e. from September to May.

Israel is divided into three ($c=3$) main areas which were in turn divided into smaller areas: The coastal area with $r_1=4$ smaller areas, the mountain area with $r_2=7$ areas and the Jordan valley with $r_3=4$ smaller areas. In each area there are several stations at which the rainfall was measured. The reader is referred to Gabriel (1972) for the actual data. It was assumed that the covariance matrices of the three main areas are equal and unknown. By using the theory in Section (2.1) the following table was constructed: The (ij) -th entry in the table represents the number of observations from the i -th area that has been classified into the j -th area.

TABLE 4.1

*Table for the Misclassification of Stations According to the Rainfall Data

Areas : J	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
I															
1	1	1													
2		4	1												
3			4												
4				4											
5					3	1									
6						3									
7							3								
8								3							
9						1	1		2						
10	1							1		3		1			
11											6				
12												3			
13													3		
14														2	1
15															2

Let $V^{(1)}$ represent the rainfall measured during the first six months of the rain season and $V^{(2)}$ the rainfall during the last three months. By using the conditional predictive density of $V^{(2)}$ given $V^{(1)}$ it can be shown that:

$$E(v^{(2)}|Y;v^{(1)}) \simeq \mu^{(2)} + \beta(v^{(1)} - \mu^{(1)})$$

where v represents V written in one column and μ and β can be found from Section (2.1). Therefore the mean rainfall for the last three months can be predicted after the rainfall measurements for the first six months are available. These predictions are tabulated in the following table:

TABLE 4.2

A comparison of the rainfall predicted for the last three months with the actual rainfall

Station	Actual rainfall			Predicted rainfall			Station	Actual rainfall			Predicted rainfall		
	March	April	May	March	April	May		March	April	May	March	April	May
1	50,0	30,3	9,4	50,4	21,6	5,9	29	42,9	22,4	4,4	48,3	22,6	5,4
2	44,7	19,7	5,4	56,5	22,5	5,3	30	37,4	19,7	3,7	52,2	21,5	4,8
3	39,7	18,4	4,2	47,5	16,1	3,8	31	66,3	31,9	4,9	68,1	25,7	4,7
4	44,5	23,5	8,0	51,0	18,7	4,3	32	88,9	38,1	3,1	86,5	25,2	4,2
5	58,5	23,0	7,5	66,5	21,6	4,4	33	56,3	20,0	3,8	41,9	18,6	4,3
6	26,8	18,8	3,6	32,9	13,0	3,8	34	82,2	28,6	2,5	72,8	25,4	4,0
7	40,4	19,2	3,8	50,2	17,4	4,2	35	69,6	24,8	4,0	71,5	24,8	4,4
8	37,9	14,1	4,2	47,3	13,0	3,1	36	54,3	19,1	2,7	48,5	24,3	5,6
9	34,4	13,8	2,3	37,2	11,2	2,6	37	68,4	22,6	2,7	54,4	20,6	4,9
10	42,4	15,8	2,1	50,6	16,3	2,8	38	57,6	14,0	2,7	47,0	17,3	5,2
11	43,8	15,0	2,5	41,7	13,7	2,9	39	38,0	11,5	1,2	36,8	8,2	2,2
12	43,0	12,0	1,5	40,0	11,6	2,6	40	21,0	5,8	1,4	19,7	8,8	2,5
13	37,4	8,5	1,4	32,3	9,8	2,7	41	30,5	7,4	4,3	22,8	8,4	2,5
14	32,9	12,6	3,5	30,0	12,2	2,9	42	15,6	5,3	4,9	17,3	5,1	2,3
15	19,8	12,3	2,6	20,3	10,1	3,0	43	19,3	5,0	3,0	14,3	6,7	2,7
16	88,4	48,4	13,9	93,6	44,7	9,9	44	15,1	7,7	3,8	10,1	6,5	2,9
17	54,5	26,9	6,5	67,1	37,5	8,9	45	62,7	35,8	11,2	64,5	31,5	9,1
18	70,5	36,8	7,9	78,4	39,3	8,6	46	53,9	25,4	11,7	61,1	24,7	8,2
19	72,0	35,6	15,9	83,6	31,8	7,3	47	46,7	24,4	8,7	55,8	22,7	6,9
20	43,4	18,8	10,2	47,9	25,3	6,3	48	38,3	22,8	5,9	47,2	19,4	5,4
21	63,3	27,5	6,4	66,6	23,1	4,9	49	42,8	22,3	5,5	44,9	21,5	5,6
22	49,0	22,4	4,1	56,7	23,6	5,8	50	27,5	12,7	9,8	30,1	16,8	5,3
23	41,0	17,4	4,9	45,6	19,6	4,4	51	17,0	7,1	3,2	14,7	7,4	1,9
24	48,9	19,4	5,2	55,2	20,4	3,7	52	11,1	4,5	1,3	7,4	4,8	1,9
25	41,4	21,5	5,2	46,5	18,4	3,8	53	6,8	3,2	1,8	5,5	3,4	1,8
26	39,6	16,7	5,0	42,2	16,9	3,8	54	11,5	2,3	4,4	8,3	3,2	1,3
27	40,3	17,7	4,2	43,5	16,3	3,7	55	4,0	4,3	1,4	3,5	3,2	2,4
28	33,2	14,4	3,2	32,8	14,2	3,8							

5. Note

In this paper details have been given only of the most important cases. Additional assumptions on the parameters have been considered in detail in Nagel and de Waal (1977 a; 1977 b).

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